

Skills Summary

Composing Transformations and Reading Symmetry

Use this reference while completing your worksheet or when studying.

Skill 1 — Combining two translations

Rule: translations add. Following $(x + a, y + b)$ by $(x + c, y + d)$ gives the single translation $(x + a + c, y + b + d)$.

The order does not matter here — two slides give the same result either way, which is not true once a reflection or rotation joins in.

Example: Translate by $(x + 4, y + 2)$, then by $(x - 9, y + 6)$. Find the single translation.
Step 1. Both steps are slides, so nothing turns or flips — the components simply add, one direction at a time.

Step 2. horizontal: $4 + (-9)$ and vertical: $2 + 6$
 $\Rightarrow (x - 5, y + 8)$

Try it: Translate by $(x - 3, y)$, then by $(x + 11, y)$. Give the horizontal component of the single translation.

check: 8

Skill 2 — Symmetry of a regular polygon

Rules: a regular polygon with n sides has n lines of symmetry, and it has rotational symmetry of order n about its centre.

The smallest rotation that maps it onto itself is $\frac{360^\circ}{n}$, and every multiple of that angle works too. Read the relation backwards to go from a given angle to n .

Example: Find the smallest rotation mapping a regular octagon onto itself.

Step 1. The eight vertices are equally spaced around one full turn, so the smallest turn that lands vertex on vertex is one eighth of that turn.

Step 2. $360^\circ / 8$
 $\Rightarrow 45^\circ$

Try it: A regular polygon's smallest rotation is 30° . Find its number of sides.

check: 12

Skill 3 — Two reflections in a row

Parallel mirrors: the result is a translation perpendicular to them, through *twice* the distance between the lines.

Intersecting mirrors: the result is a rotation about the crossing point, through *twice* the angle between the lines.

The doubling is the whole point — reflecting back and forth covers the gap twice.

Example: Reflect in $x = 2$, then in $x = 7$. Find the translation distance.

Step 1. The two mirrors are parallel, so the composition is a translation, and the rule doubles the gap between the lines rather than using it once.

Step 2. $2(7 - 2) = 2(5)$
 $\Rightarrow$ 10 units

Try it: Two mirrors through the origin meet at 40° . Find the angle of the resulting rotation.

check: 80°

Skill 4 — Naming the single equivalent motion

Method: apply the composition to one point, compare the image with the original, and match the pattern to a rule.

both coordinates negated $\Rightarrow$ half turn about the origin y negated only $\Rightarrow$ reflection in the x -axis x negated only $\Rightarrow$ reflection in the y -axis both shifted by fixed amounts $\Rightarrow$ translation

Example: Reflect $(7, 3)$ in the x -axis, then in the y -axis. Name the single motion.

Step 1. Tracking one point is enough to identify the pattern, so apply both steps to it and then compare the result with where it started.

Step 2. $(7, 3) \rightarrow (7, -3) \rightarrow (-7, -3)$: both coordinates have changed sign
 $\Rightarrow$ (-7 , -3), which is a half turn about the origin.

Try it: Reflect $(4, -9)$ in the x -axis. Give the y -coordinate of the image.

check: 6